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Support_Ticket_Classification (1).ipynbSupport_Ticket_Classification.ipynbwebsite.HTML

C: > Users > divya > Downloads > Support_Ticket_Classification (1).ipynb > print()

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print(df.shape)

df.head()

Python

(200000, 30)

	ticket_id	customer_name	customer_email	product	category	issue_description	resolution_notes	priority	status	channel	...	ticket_resolved_date	escalated	sla_breached
0	1	Patricia Smith	patricia.smith760@outlook.com	Web Portal	Account Suspension	The payment was deducted from my bank account ...	Data synchronization restored after backend se...	Urgent	Open	Email	...	2023-05-20	No	Yes
1	2	Patricia Williams	patricia.williams390@gmail.com	Mobile App	Performance Issue	I found a bug in the latest update affecting r...	Provided step-by-step troubleshooting instruct...	Urgent	Closed	Email	...	2024-01-19	Yes	Yes
2	3	William Anderson	william.anderson651@outlook.com	Web Portal	Performance Issue	The application crashes whenever I try to uplo...	Provided step-by-step troubleshooting instruct...	Medium	Closed	Chat	...	2022-12-05	Yes	Yes
3	4	David Miller	david.miller672@icloud.com	Payment Gateway	Subscription Cancellation	My subscription was cancelled without my reque...	Provided step-by-step troubleshooting instruct...	Medium	Closed	Social Media	...	2024-04-04	Yes	No
4	5	Robert Gonzalez	robert.gonzalez391@hotmail.com	Web Portal	Feature Request	The system is not syncing data across devices ...	We have reset the account credentials and advi...	High	Pending Customer	Email	...	2024-08-24	Yes	No

5 rows x 30 columns

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C:\Users>divya>Downloads>Support_Ticket_Classification (1).ipynb>print(df.columns)

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print(df.columns)

Python

Index(['ticket_id', 'customer_name', 'customer_email', 'product', 'category',
'issue_description', 'resolution_notes', 'priority', 'status',
'channel', 'region', 'customer_age', 'customer_gender',
'subscription_type', 'customer_tenure_months', 'previous_tickets',
'customer_satisfaction_score', 'first_response_time_hours',
'resolution_time_hours', 'ticket_created_date', 'ticket_resolved_date',
'escalated', 'sla_breached', 'operating_system', 'browser',
'payment_method', 'language', 'preferred_contact_time',
'issue_complexity_score', 'customer_segment'],
dtype='object')

df = df[['issue_description', 'category', 'priority']]

Python

print(df.isnull().sum())

Python

issue_description0
category0
priority0
dtype: int64

df = df.dropna()

Python

13

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C:\Users>divya>Downloads>Support_Ticket_Classification (1).ipynb>df[['issue_description','cleaned_text']].head()

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Select Kernel

df[['issue_description','cleaned_text']].head()

[10]Python

	issue_description	cleaned_text
0	The payment was deducted from my bank account ...	the payment was deducted from my bank account ...
1	I found a bug in the latest update affecting r...	i found a bug in the latest update affecting r...
2	The application crashes whenever I try to uplo...	the application crashes whenever i try to uplo...
3	My subscription was cancelled without my reque...	my subscription was cancelled without my reque...
4	The system is not syncing data across devices ...	the system is not syncing data across devices ...

x = df[['cleaned text']]

y = df[['category']]

[11]Python

vectorizer = TfidfVectorizer(
 max_features=5000,
 stop_words='english'
)

x_tfidf = vectorizer.fit_transform(x)

[12]Python

x_train, x_test, y_train, y_test = train_test_split(
 x_tfidf,

[13]Python

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C:\> Users > divya > Downloads > Support_Ticket_Classification (1).ipynb > model = LogisticRegression()

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Select Kernel

```
X_train, X_test, y_train, y_test = train_test_split(
    X_train,
    y_train,
    test_size=0.2,
    random_state=42
)
```

[13]Python

```
model = LogisticRegression(
    max_iter=1000
)

model.fit(
    X_train,
    y_train
)
```

Python

LogisticRegression

LogisticRegression(max_iter=1000)

```
y_pred = model.predict(X_test)
```

[15]Python

```
accuracy = accuracy_score(
    y_test,
    y_pred
)
```

[17]Python

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[17]
)print("Accuracy -", accuracy)
Python

...Accuracy - 0.097475

[20]
print(
 classification_report(y_test,y_pred,zero_division=0)
)
Python

...

	precision	recall	f1-score	support
Account Suspension	0.10	0.10	0.10	3973
Bug Report	0.00	0.00	0.00	4009
Data Sync Issue	0.00	0.00	0.00	4013
Feature Request	0.09	0.19	0.13	3941
Login Issue	0.00	0.00	0.00	4022
Payment Problem	0.09	0.09	0.09	4006
Performance Issue	0.09	0.10	0.10	3967
Refund Request	0.10	0.19	0.13	3983
Security Concern	0.10	0.20	0.13	3949
Subscription Cancellation	0.11	0.11	0.11	4137
accuracy			0.10	40000
macro avg	0.07	0.10	0.08	40000
weighted avg	0.07	0.10	0.08	40000

[21]
print(

Python

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weighted avg0.070.100.0840000

```
print(  
    confusion_matrix(  
        y_test,  
        y_pred  
    )  
)
```

[21]Python

[[384 0 0 779 0 415 414 789 787 405]
[373 0 0 835 0 419 421 822 781 358]
[393 0 0 872 0 410 393 746 780 419]
[385 0 0 765 0 376 405 811 793 406]
[438 0 0 796 0 385 389 787 795 432]
[435 0 0 810 0 373 411 778 809 390]
[382 0 0 803 0 423 383 766 822 388]
[413 0 0 806 0 391 391 776 812 394]
[410 0 0 758 0 376 422 806 781 396]
[429 0 0 841 0 382 414 782 852 437]]

```
x_priority = df['cleaned_text']  
y_priority = df['priority']
```

[22]Python

```
x_priority_tfidf = vectorizer.fit_transform(  
    x_priority  
)
```

[23]Python

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Select Kernel

priority_model.fit(
 Xp_train,
 yp_train
)

[25]Python

LogisticRegression
LogisticRegression(max_iter=1000)

priority_pred = priority_model.predict(
 Xp_test
)

[26]Python

print(
 accuracy_score(
 yp_test,
 priority_pred
)
)

[27]Python

0.24955

print(
 classification_report(
 yp_test,
 priority_pred,
 display_names=False
)
)

[29]Python

Hot weather
Now

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[27] Python
... 0.24955

```
print(
  classification_report(
    yp_test,
    priority_pred, zero_division=0
  )
)
```

[28] Python

	precision	recall	f1-score	support
High	0.26	0.30	0.28	10152
Low	0.25	0.20	0.22	9885
Medium	0.00	0.00	0.00	10062
Urgent	0.25	0.50	0.33	9901
accuracy			0.25	40000
macro avg	0.19	0.25	0.21	40000
weighted avg	0.19	0.25	0.21	40000

```
new_ticket = [
  "Payment failed during checkout process"
]
```

Python

```
clean_ticket = clean_text(
```




